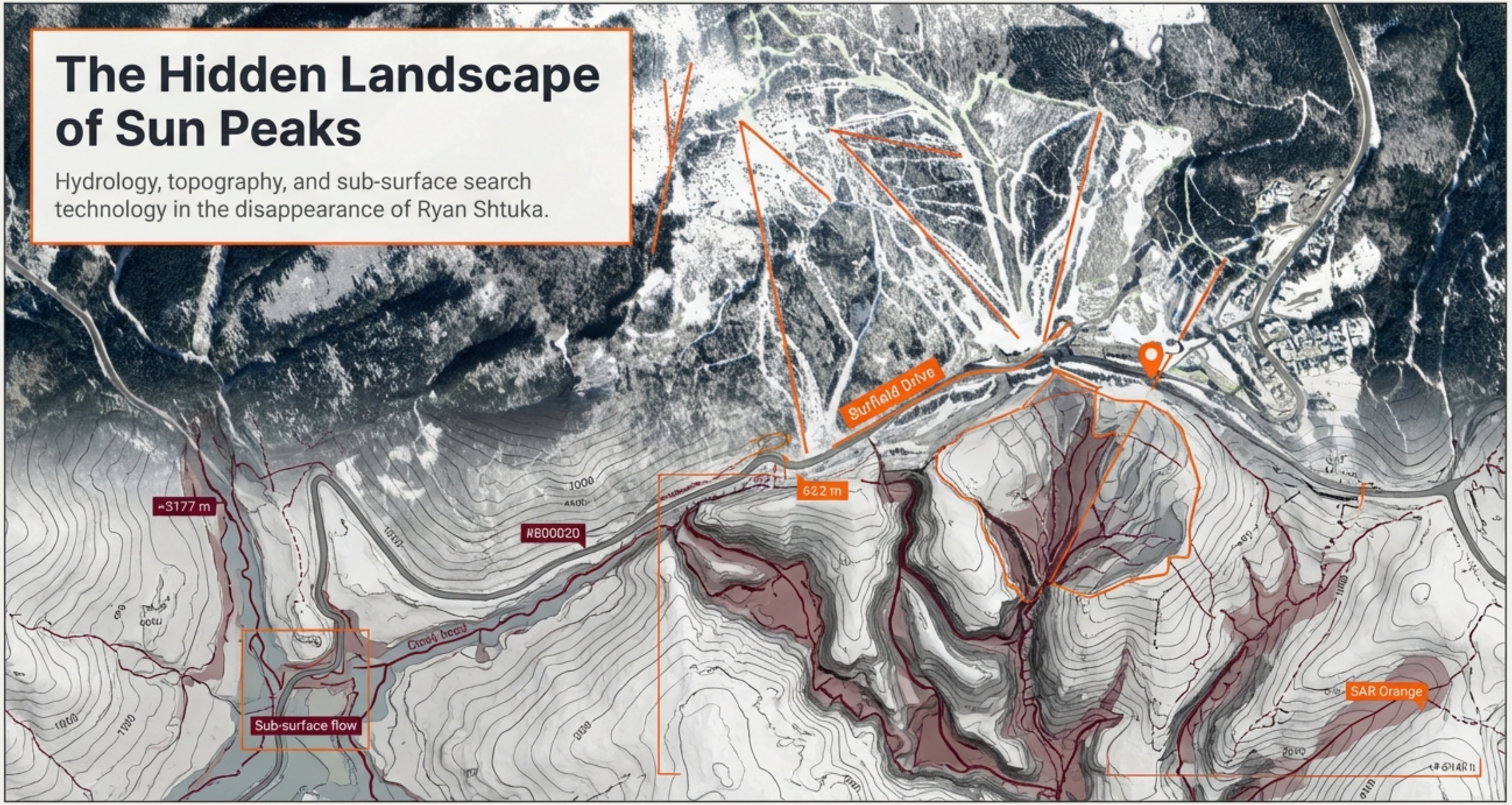
The Hidden Landscape of Sun Peaks

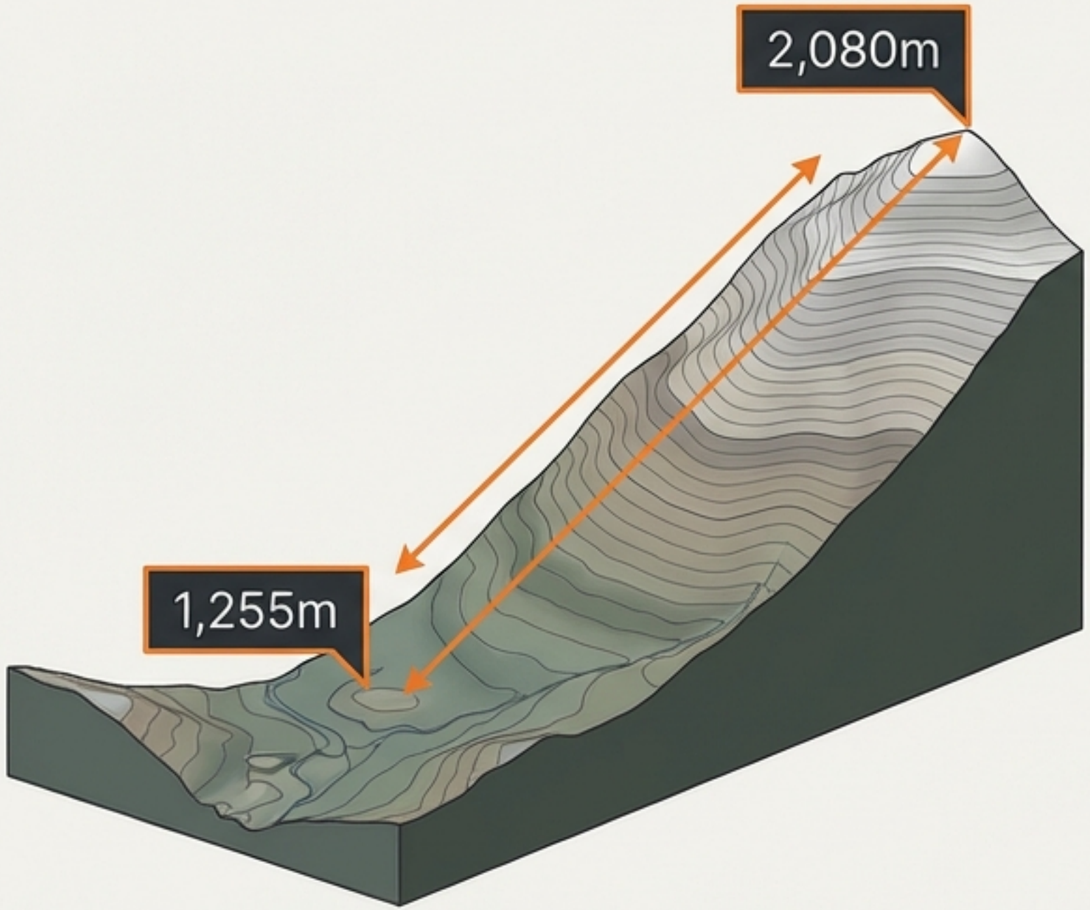
Hydrology, topography, and sub-surface search technology in the disappearance of Ryan Shtuka.

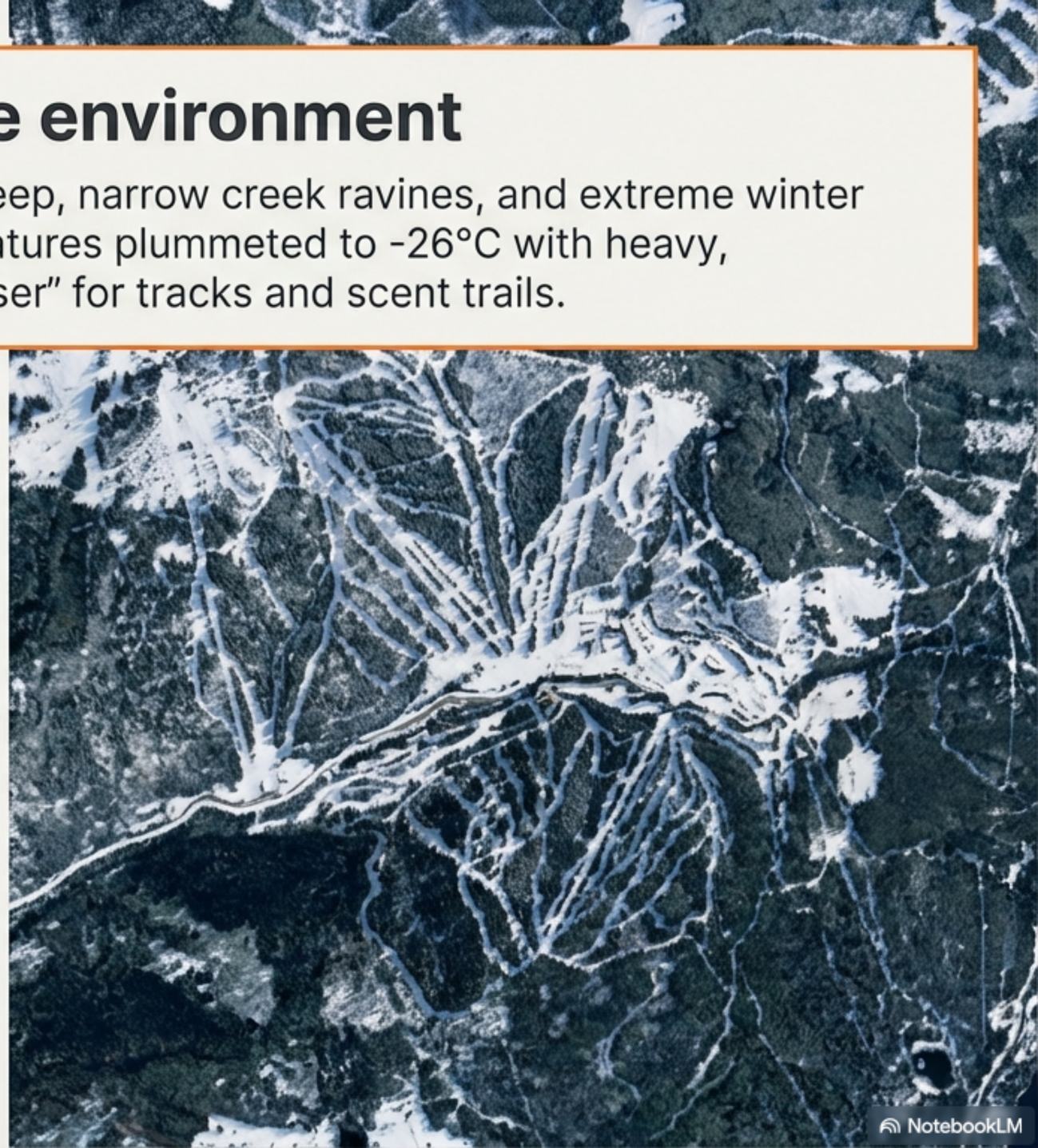


NotebookLM

A deceptively rugged alpine environment

Sun Peaks is characterized by dense forests, steep, narrow creek ravines, and extreme winter conditions. On the night Ryan vanished, temperatures plummeted to -26°C with heavy, continuous snowfall that acted as a “natural eraser” for tracks and scent trails.

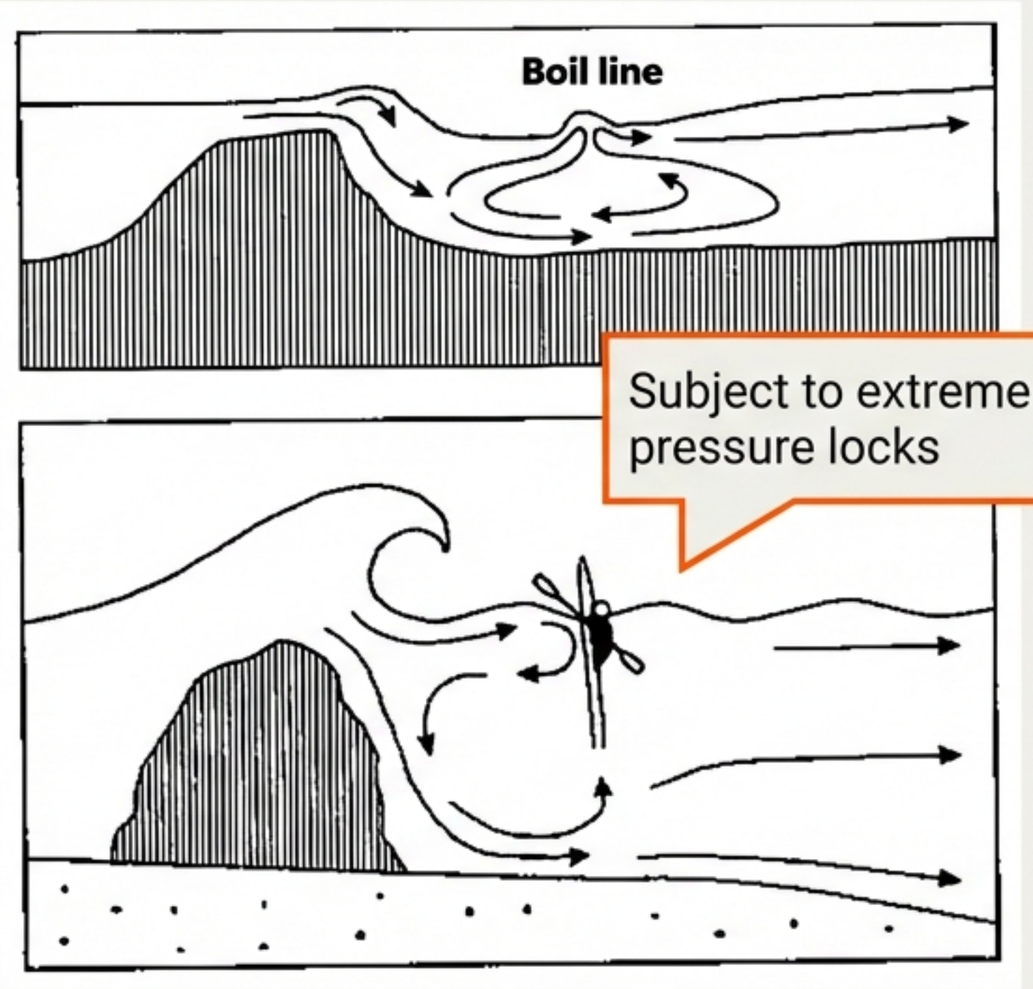




NotebookLM

Understanding hydrological ‘Strainers’

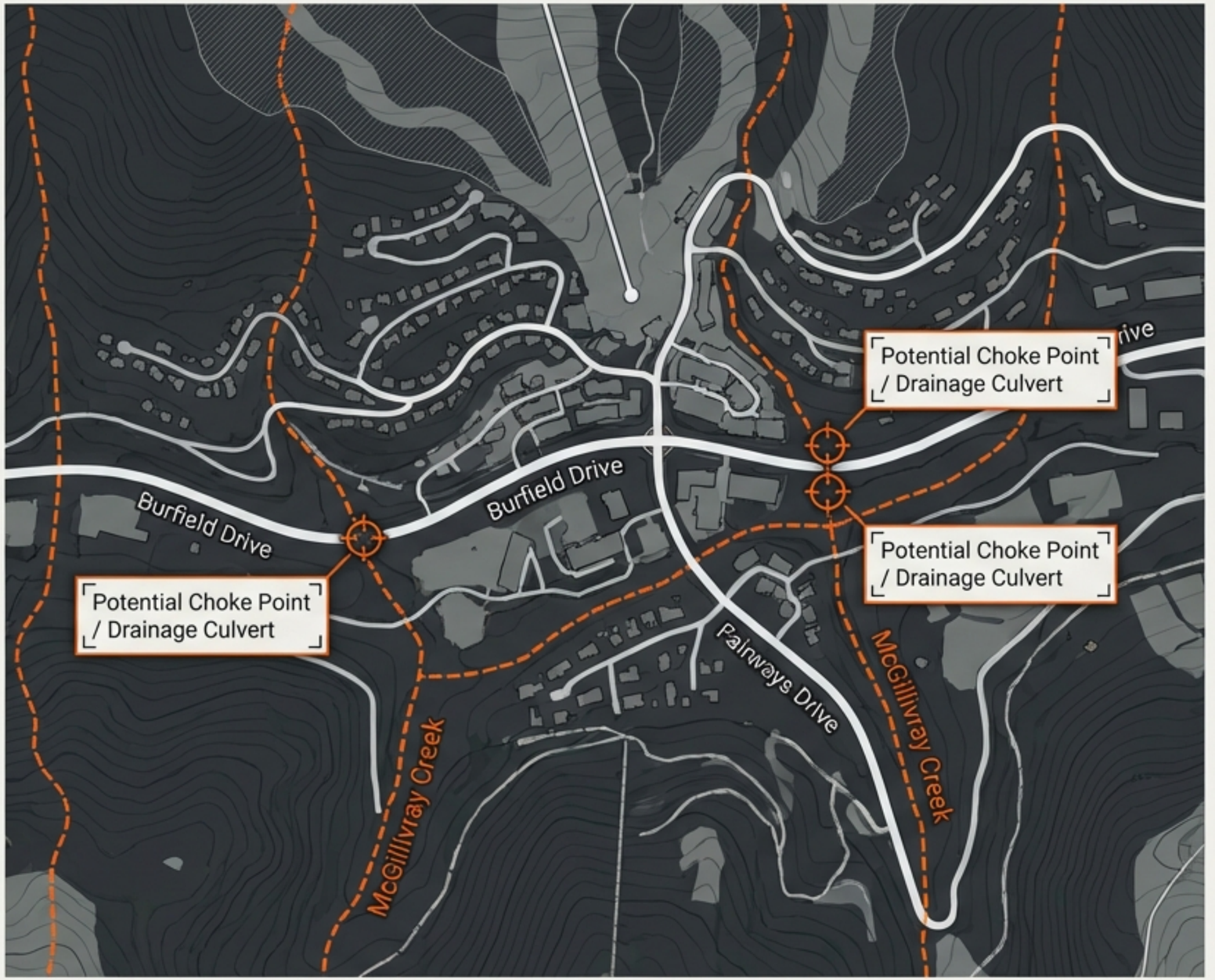
In swiftwater SAR, a strainer is solid material in the water—like fallen trees, boulders, fences, or grates—that allows water to flow through but traps solid objects. The immense force of the water creates a pressure lock, making it nearly impossible to dislodge an object trapped against it.



NotebookLM

Natural choke points along Burfield Drive

Strainers naturally form in the steep, densely forested creek ravines cutting through Sun Peaks. Prime hazard zones exist where McGillivray Creek narrows, where deadfall accumulates, and where water is forced under modern roads via drainage culverts.



NotebookLM

The limits of seasonal surface searches

Initial searches by RCMP dive teams, helicopters, and 74 volunteers were extensive. However, spring thaws and rapid creek flows drastically alter the landscape. Over seven years, snowmelt pushes debris deeper, burying objects under years of new foliage, ie, deadfall, and shifting creek beds.

FEBRUARY 2018: Static frozen landscape.



APRIL 2019 ONWARD: Dynamic shifting hazards.

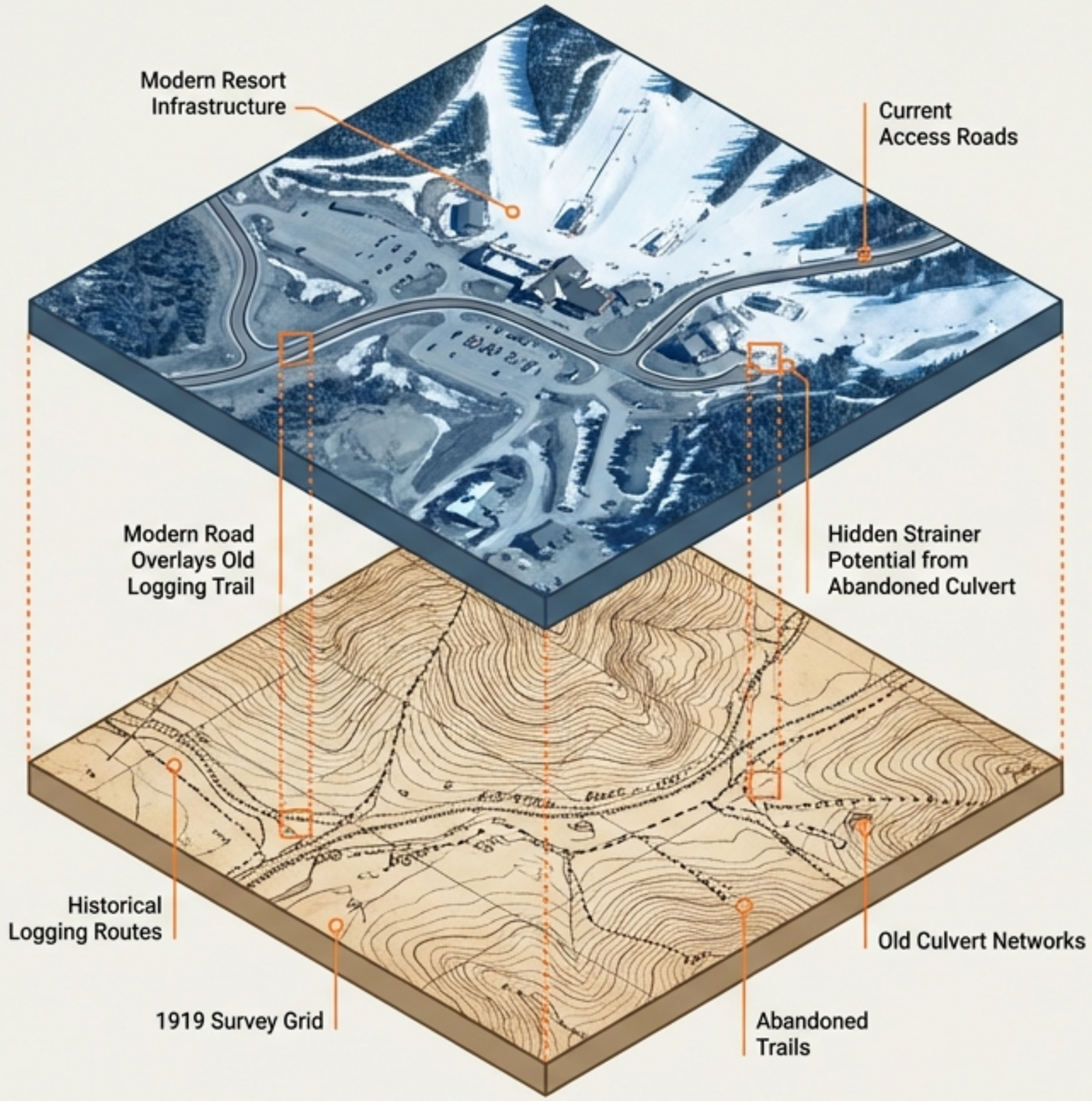


Question: Were water hazards checked? Answer: Yes, but spring thaws constantly create and bury new strainers.

NotebookLM

The threat of undocumented logging infrastructure

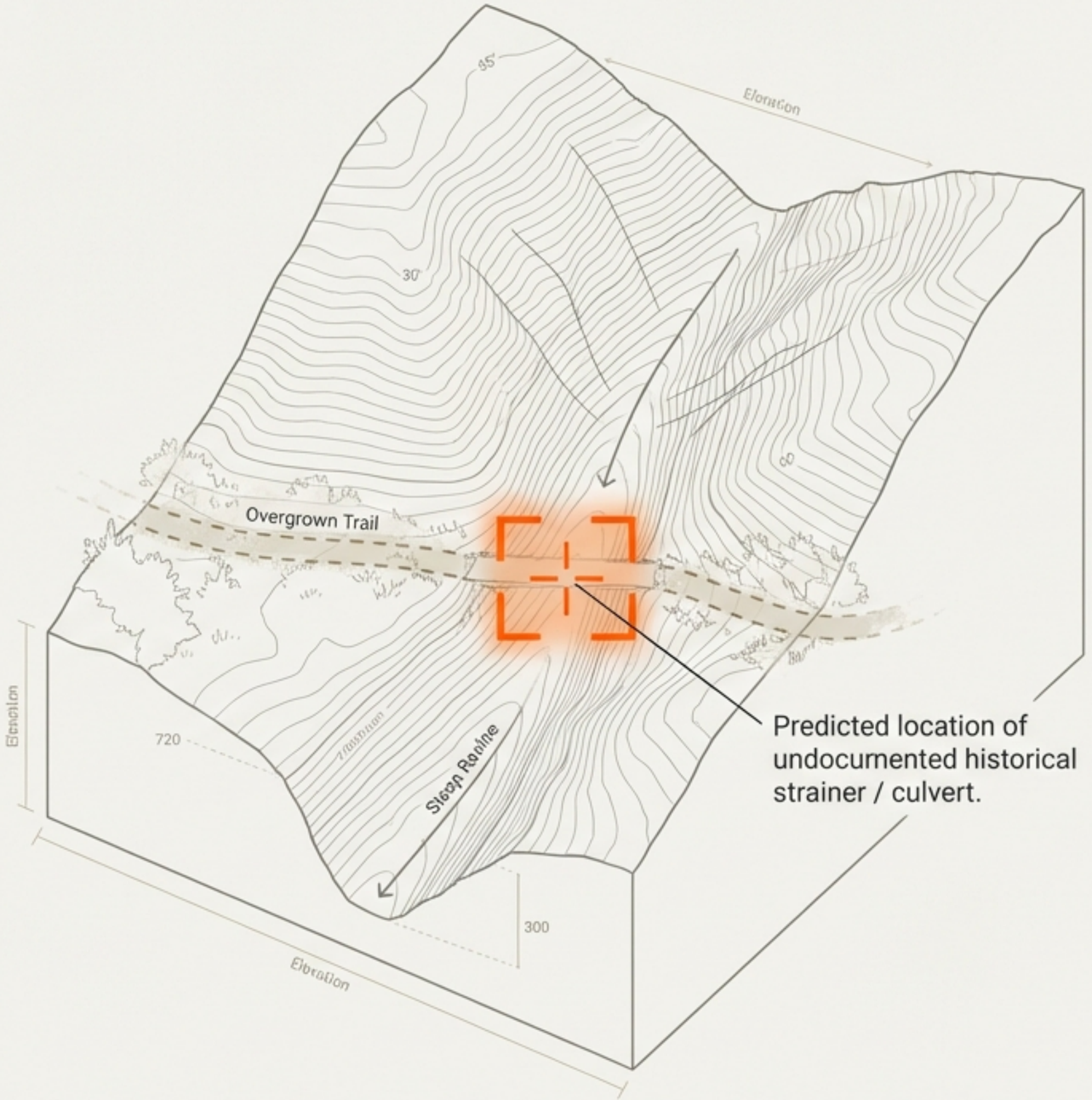
The Kamloops and Thompson Okanagan regions have a rich history of resource extraction, thoroughly documented in the 1919–20 Department of the Interior topographical surveys. The mountainsides around the resort are layered with forgotten logging roads, overgrown access trails, and abandoned drainage culverts that function as hidden strainers.



NotebookLM

Predicting the locations of hidden traps

Finding undocumented hazards requires forensic geography. By cross-referencing modern satellite data with historical logging maps, analysts can identify ‘terrain funnels.’
Where steep ravines intersect with old trails, makeshift bridges or undocumented drainage pipes were likely installed.



NotebookLM

Targeted drone photogrammetry in Ryan’s case

Aerial mapping was conducted privately by drone operator Ryan Hillaby, who dedicated a 7-day flight week to photograph hard-to-reach ravines.

The high-resolution imagery was processed using software by Shane Michaels (Wings of Mercy) specifically calibrated to scan the landscape for the color of Ryan’s burgundy ball cap.



NotebookLM

Penetrating the canopy with LiDAR

While drones capture surface colors, Light Detection and Ranging (LiDAR) uses laser pulses to penetrate dense forest canopies and leaf litter. Utilizing high-resolution data from Natural Resources Canada is the premier method for exposing overgrown logging infrastructure, ground disturbances, and hidden topographical features.

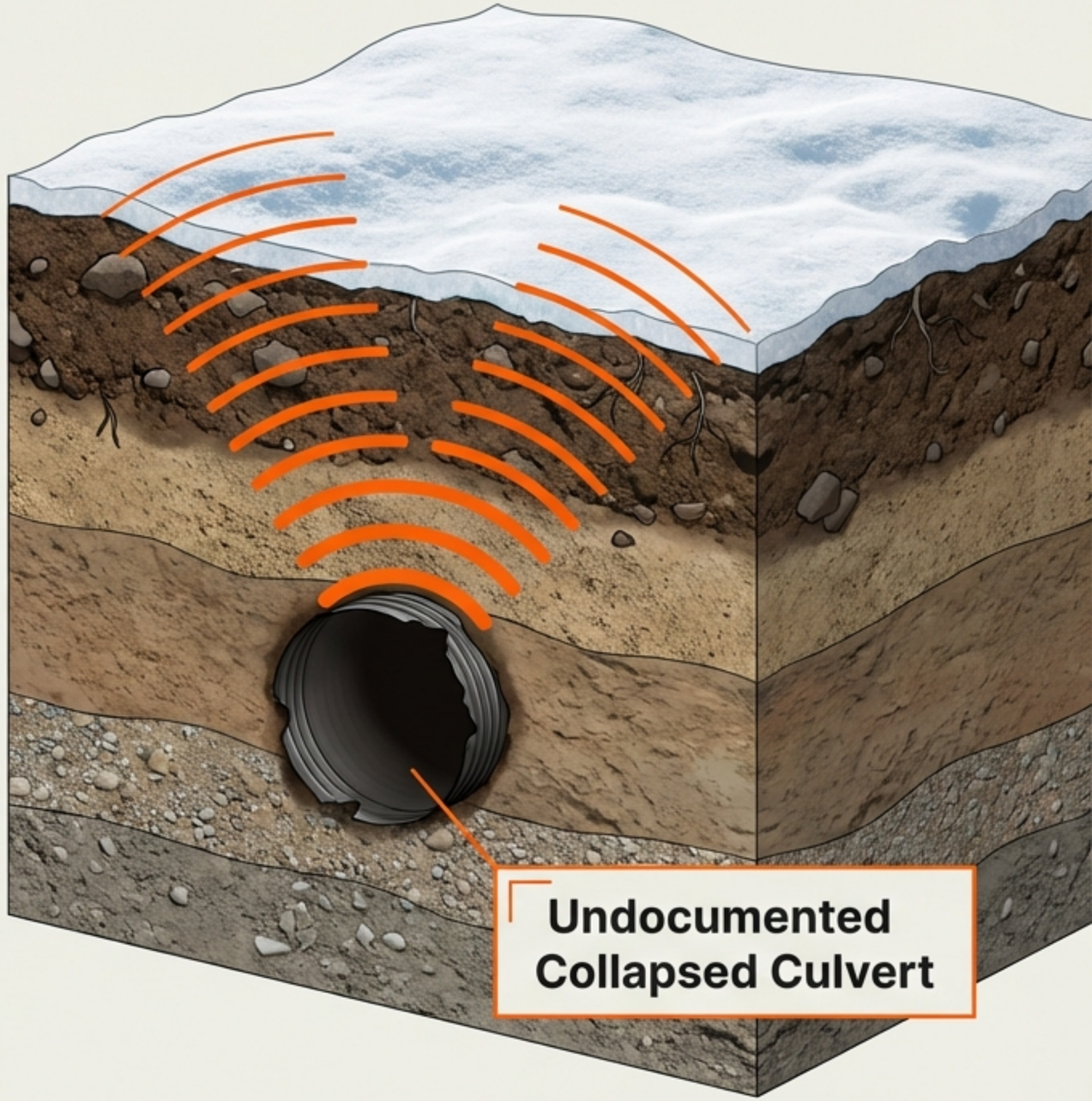


NotebookLM

Sub-surface searching with Ground Penetrating Radar

When LiDAR maps the surface, Ground Penetrating Radar (GPR) looks beneath it. GPR sends radar pulses into the earth to find anomalies under soil and debris.

SAR experts identify this as the vital technology needed to search collapsed culverts and areas where debris has piled up for years.

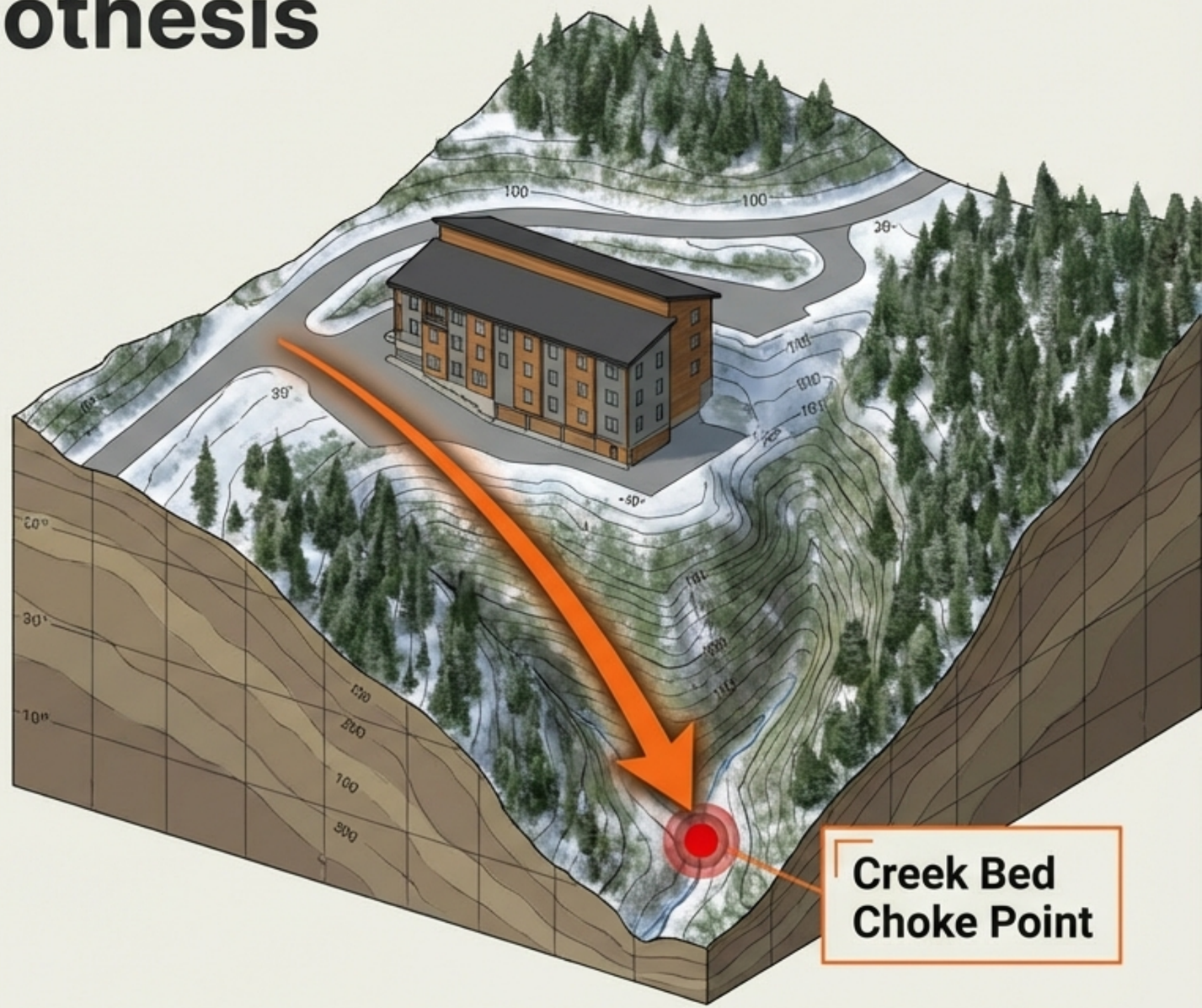


NotebookLM

The ‘Trap Zone’ Hypothesis

In -26°C temperatures, a disoriented individual leaving Burfield Drive could easily slip into a steep, snow-covered ravine.

The natural topography would act as a funnel, directing a fall or slide straight into a creek bed and toward a choke point—either a natural deadfall strainer or an undocumented logging culvert.



NotebookLM

Actionable next steps for search operations



Map the Waterways: Trace the exact hydrological path of McGillivray Creek downstream from Burfield Drive.



Locate the Choke Points: Cross-reference historic maps to identify every natural strainer, modern grate, and undocumented logging culvert.



Deploy Sub-Surface Tech: Utilize GPR on collapsed culverts and analyze NRCan bare-earth LiDAR to expose hidden infrastructure in the ravines.

NotebookLM